

Mathematics for Computer Science

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Chapter 1

Recursion

1.1 Tower of Hanoi

Recursive Formula

Let $T(n)$ denote the minimum number of moves required to solve the Tower of Hanoi problem with n disks.

Recurrence relation	Single Tower of Hanoi	Double Tower of Hanoi	Triple Tower of Hanoi
	: $T_n = 2T_{n-1} + 1$	: $T_n = 2T_{n-1} + 2$	: $T_n = 2T_{n-1} + 3$

Close Form	Single Tower of Hanoi	Double Tower of Hanoi	Triple Tower of Hanoi
	: $T_n = 1 (2^n - 1)$	: $T_n = 2 (2^n - 1)$	: $T_n = 3 (2^n - 1)$

Find the number moves required to move 8 disks

Recursion	↓	$T_8 = 2T_7 + 1 = 2 \times 127 + 1 = 255$	↑	Back calculation
		$T_7 = 2T_6 + 1 = 2 \times 63 + 1 = 127$		
		$T_6 = 2T_5 + 1 = 2 \times 31 + 1 = 63$		
		$T_5 = 2T_4 + 1 = 2 \times 15 + 1 = 31$		
		$T_4 = 2T_3 + 1 = 2 \times 7 + 1 = 15$		
		$T_3 = 2T_2 + 1 = 2 \times 3 + 1 = 7$		
		$T_2 = 2T_1 + 1 = 2 \times 1 + 1 = 3$		
		$T_1 = 1$		

For $n = 8$ disks, $T(8) = 2^8 - 1 = 256 - 1 = 255$

finding recursive formula

$$n=1 \quad \downarrow \uparrow \rightsquigarrow ||$$

$$T(1) = 1$$

$$n=2 \quad \downarrow \uparrow \rightsquigarrow \downarrow \uparrow \rightsquigarrow | \downarrow \uparrow \rightsquigarrow ||$$

$$T(2) = 3 = 2 \cdot T(1) + 1$$

$$n=3 \quad \downarrow \uparrow \rightsquigarrow \downarrow \uparrow \rightsquigarrow | \downarrow \uparrow \rightsquigarrow || \quad T(3) = 2 \cdot T(2) + 1$$

$$= 2 \cdot 3 + 1$$

$$= 7$$

$$n=4 \quad \downarrow \uparrow \rightsquigarrow \downarrow \uparrow \rightsquigarrow | \downarrow \uparrow \rightsquigarrow || \quad T(4) = 2 \cdot T(3) + 1$$

$$= 2 \cdot 7 + 1$$

$$= 15$$

recursive formula:

$$\boxed{\begin{array}{l} T(n) = 2 \cdot T(n-1) + 1 \\ T(1) = 1 \end{array}}$$

Closed Form

From the recurrence relation of TOH,

$$\begin{aligned}
 T_n &= 2T_{(n-1)} + 1 \\
 &= 2(2T_{(n-2)} + 1) + 1 \\
 &= 2^2T_{(n-2)} + 2 + 1 \\
 &= 2^2(2T_{(n-3)} + 1) + 2 + 1 \\
 &= 2^3T_{n-3} + 2^2 + 2 + 1 \\
 &\vdots \\
 &= 2^nT_{n-n} + (2^{n-1} + 2^{n-2} + \dots + 2^1 + 2^0) \\
 &= 0 + \sum_{k=0}^{n-1} 2^k \\
 &= 2^{(n-1)+1} - 1 \quad \left[\because \sum_{k=0}^n 2^k = 2^{n+1} - 1 \right] \\
 \therefore T_n &= 2^n - 1
 \end{aligned}$$

From the recurrence relation of DTOH,

$$\begin{aligned}
 T_n &= 2T_{(n-1)} + 2 \\
 &= 2(2T_{(n-2)} + 2) + 2 \\
 &= 2^2T_{(n-2)} + 2 \cdot 2 + 2 \\
 &= 2^2(2T_{(n-3)} + 2) + 2 \cdot 2 + 2 \\
 &= 2^3T_{n-3} + 2^2 \cdot 2 + 2^1 \cdot 2 + 2^0 \cdot 2 \\
 &= 2^3T_{n-3} + 2(2^2 + 2^1 + 2^0) \\
 &\vdots \\
 &= 2^nT_{n-n} + 2(2^{n-1} + 2^{n-2} + \dots + 2^1 + 2^0) \\
 &= 0 + 2 \sum_{k=0}^{n-1} 2^k \\
 &= 2(2^{(n-1)+1} - 1) \\
 \therefore T_n &= 2(2^n - 1)
 \end{aligned}$$

Proof by Induction

Close form proof of TOH: $T_n = (2^n - 1)$,

Base Case:

For the lowest value of $n = 1$,

$$T_1 = 2^1 - 1 = 2 - 1 = 1$$

Hypothesis:

Assume the formula holds for $n = 1$ to $k - 1$,

$$T_{k-1} = 2^{k-1} - 1$$

Inductive Step:

From the recurrence,

$$\begin{aligned}
 T_k &= 2T_{(k-1)} + 1 \\
 &= 2(2^{k-1} - 1) + 1 \\
 &= 2^{k-1+1} - 2 + 1 \\
 &= 2^k - 1
 \end{aligned}$$

Thus, the formula also holds for $n = k$. (Proved)

Close form proof of DTOH: $T_n = 2(2^n - 1)$,

Base Case:

For the lowest value of $n = 1$,

$$T_1 = 2(2^1 - 1) = 2(2 - 1) = 2$$

Hypothesis:

Assume the formula holds for $n = 1$ to $k - 1$,

$$T_{k-1} = 2(2^{k-1} - 1)$$

Inductive Step:

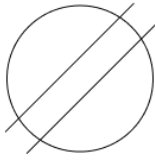
From the recurrence,

$$\begin{aligned}
 T_k &= 2T_{(k-1)} + 2 \\
 &= 2 \cdot 2(2^{k-1} - 1) + 2 \\
 &= 2[2(2^{k-1} - 1) + 1] \\
 &= 2[2^{k-1+1} - 2 + 1] \\
 &= 2(2^k - 2)
 \end{aligned}$$

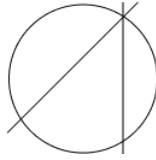
Thus, the formula also holds for $n = k$. (Proved)

1.2 Lines in the plane

Rules:

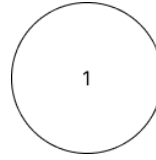


Lines can't be
parallel to each other

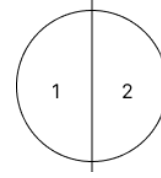


Lines can't
intersect in border

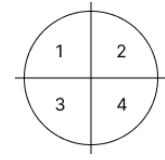
Regions



$$L_0 = 1$$



$$L_1 = 2$$



$$L_2 = 4$$

$\therefore$ Recursive formula: $L_n = L_{n-1} + n$

Close Form

$$\begin{aligned}
 L_n &= L_{n-1} + n \\
 &= L_{n-2} + (n-1) + n & [\because L_{n-1} &= L_{n-2} + (n-1)] \\
 &= L_{n-3} + (n-2) + (n-1) + n & [\because L_{n-2} &= L_{n-3} + (n-2)] \\
 &\vdots \\
 &= L_{n-n} + \cdots + (n-2) + (n-1) + n \\
 &= L_0 + \sum_{k=1}^n k \\
 &= 1 + \frac{n(n+1)}{2}
 \end{aligned}$$

Proof by Induction

Proof that n lines will divide a plane into $L_n = 1 + \frac{n(n+1)}{2}$ regions,

Base Case:

For the lowest value of $n = 0$,

$$L_0 = 1 + \frac{0(0+1)}{2} = 1 + 0 = 1$$

Hypothesis:

Assume the formula holds for $n = 1$ to $n - 1$,

$$L_{n-1} = 1 + \frac{(n-1)(n-1+1)}{2} = 1 + \frac{n(n-1)}{2}$$

Inductive Step:

From the recurrence,

$$\begin{aligned}
 L_n &= L_{n-1} + n = 1 + \frac{n(n-1)}{2} + n \\
 &= 1 + \frac{n^2 - n}{2} + n = 1 + \frac{n^2 - n + 2n}{2} \\
 &= 1 + \frac{n^2 + n}{2} = 1 + \frac{n(n+1)}{2}
 \end{aligned}$$

Thus, the formula also holds for n . (Proved)

1.3 Josephus Problem

Recurrence formula $J(2n) : 2J(n) - 1$ (even)
 $J(2n + 1) : 2J(n) + 1$ (odd)

Josephus Close Form: $n = 2^m + l$ and $n_{th} = 2l + 1$

$J(n)$: gives survival position $\rightarrow 2l + 1$

n : indicates n number of persons participated $\rightarrow 2^m + l$

$J(n) \rightarrow n$ -th position survived

$J(2^m + l) \rightarrow (2l + 1)$ -th position survived

Question: Find $J(103)$

Solution: Using recursion / recurrence formula:

Recursion	$J(103) = J(2 \times 51 + 1) = 2J(51) + 1 = 79$	Back calculation
	$J(51) = J(2 \times 25 + 1) = 2J(25) + 1 = 39$	
	$J(25) = J(2 \times 12 + 1) = 2J(12) + 1 = 19$	
	$J(12) = J(2 \times 6) = 2J(6) - 1 = 9$	
	$J(6) = J(2 \times 3) = 2J(3) - 1 = 5$	
	$J(3) = J(2 \times 1 + 1) = 2J(1) + 1 = 3$	
	$J(1) = 1$	

Using close form formula:

Here, $103 = 2^6 + 39$, $\therefore l = 39$

$\therefore$ Survival position: $n = 2l + 1 = 2(39) + 1 = 79$

Using binary property:

$J(103) = ?$

$(103)_{10} = (1100111)_2$

By applying binary left rotation,

we get: $(1001111)_2 = (79)_{10}$

The Josephus Problem is a theoretical puzzle involving people standing in a circle and being eliminated in a fixed pattern until only one person remains. Find the smallest three values of N such that the person standing at the $N/3$ -rd position survives.

Proof by Induction

Induction:

Even: $T(2n) = 2T(n) - 1 \quad \text{--- (1)}$

$$2n = 2^m + l$$

$$n = 2^{m-1} + l/2$$

now in $T(n)$:

$$T(2n) = 2T(n) - 1$$

$$= 2 \left(2 \times l/2 + 1 \right) - 1$$

$$= 2l + 2 - 1$$

$$= 2l + 1$$

Odd: $T(2n+1) = 2T(n) + 1 \quad \text{--- (2)}$

$$2n+1 = 2^m + l \quad \left| \begin{array}{l} n = 2^{m-1} + \frac{l-1}{2} \end{array} \right.$$

Now in (2):

$$T(2n+1) = 2T(n) + 1$$

$$= 2 \left(2 \times \frac{l-1}{2} + 1 \right) + 1$$

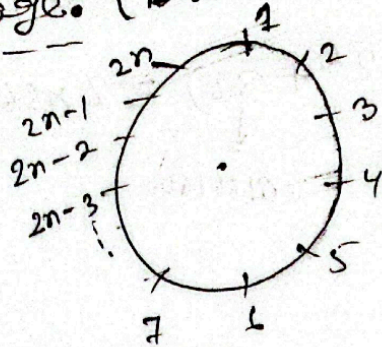
$$= 2l - 2 + 2 + 1$$

$$= 2l + 1$$

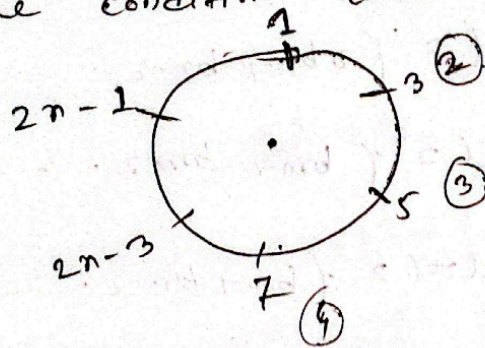
proved

Recurrence Relation

Initial stage: (Before starting the trip)



Now, after first trip start from (1),
All even numbers are eliminated
So, the condition looks like:



$$\begin{aligned} 4n-2 \\ 2n-6-1 \\ 2(2n-3)-1 \end{aligned}$$

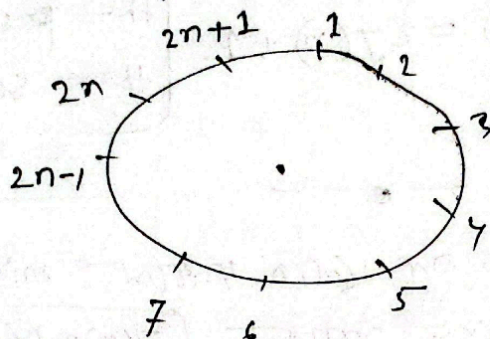
Old position	New position
1	1
2	3
3	5
5	7
⋮	⋮

So, after every trip starting with (1)
except each person has been doubled
and decremented by 1.

let, $(p = 2n) \rightarrow$ (for even index)
 we get $J(2n) = 2J(n) - 1$.

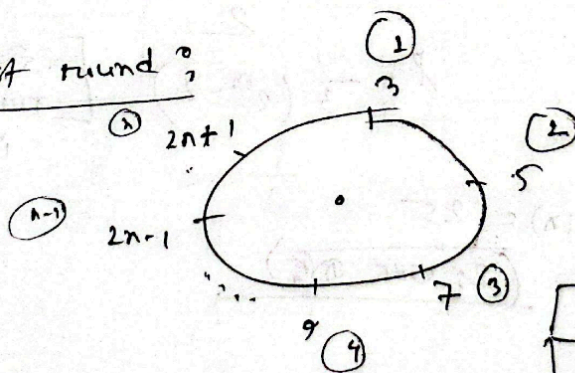
For odd case:

Initially the position of the people :



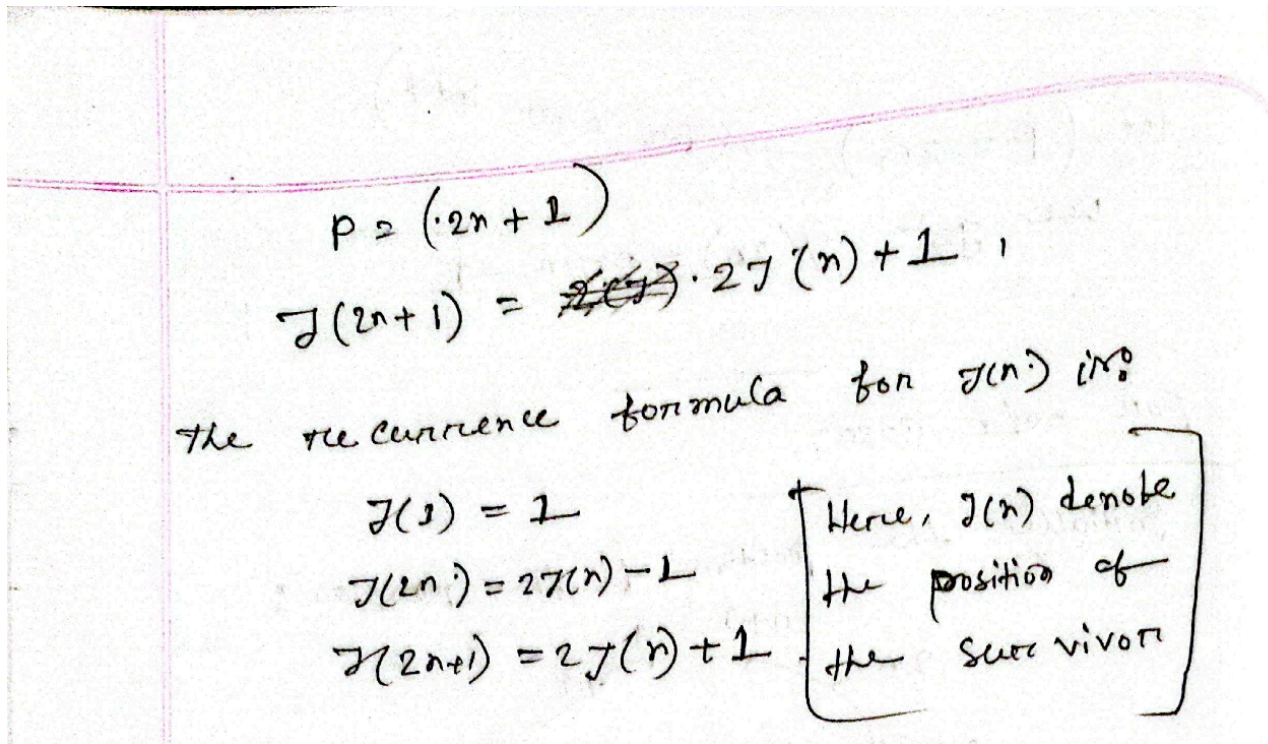
After first trip all even index are wiped out, also 1 is wiped out after $2n$.

After first round :



After every trip, starting with p except that now each person is doubled and increased by 2

old pos	new pos
1	3
2	5
3	7
⋮	⋮



Question: Find the smallest 3 Values of n where $n/3^{\text{rd}}$ person will survived.

Given,

$$J(n) = \frac{n}{3}$$

$$2l + 1 = \frac{(2^m + l)}{3}$$

$$3(2l + 1) = (2^m + l)$$

$$6l + 3 = 2^m + l$$

$$5l = 2^m - 3$$

$$\therefore l = \frac{(2^m - 3)}{5}$$

l must be an integer, so we take only those values of m where $(2^m - 3)$ is divisible by 5.

Now, $\therefore m = 3, 7, 11$

m	$l = \frac{(2^m - 3)}{5}$	$n = 2^m + l$	$J(n) = 2l + 1 = \frac{n}{3}$
3	1	9	3
7	25	153	51
11	409	2457	819

Question: Find the smallest 3 Values of n where 25^{th} person will survived.

Given, $J(n) = 25$ $2l + 1 = 25$ $\therefore l = 12$

m	l	$n = 2^m + l$	
1	12	14	(not accepted since $14 < 25$)
2	12	16	(not accepted since $16 < 25$)
3	12	20	(not accepted since $20 < 25$)
4	12	28	(accepted)
5	12	44	(accepted)
6	12	76	(accepted)

Since 25-th person survived therefore at 25 persons needs to be participated.

Hence, $n = 14, 16, 20$ are not acceptable.

Thus, smallest 3 values of $n = 28, 44, 76$

1.4 Ackermann Function

$$A(m, n) = \begin{cases} n + 1, & m = 0, \\ A(m - 1, 1), & m > 0, n = 0, \\ A(m - 1, A(m, n - 1)), & m > 0, n > 0. \end{cases}$$

Example: Compute $A(2, 2)$

$$A(2, 2) = A(1, A(2, 1)) \quad \dots (1)$$

$$A(2, 1) = A(1, A(2, 0)) \quad \dots (2)$$

$$A(2, 0) = A(1, 1) \quad \dots (3)$$

$$A(1, 1) = A(0, A(1, 0)) \quad \dots (4)$$

From (4): $A(1, 0) = A(0, 1) = 2 \Rightarrow A(1, 1) = A(0, 2) = 3$.

From (3): $A(2, 0) = A(1, 1) = 3$.

From (2): $A(2, 1) = A(1, 3)$.

Now, $A(1, 3) = A(0, A(1, 2))$, $\dots (5)$

From (5): $A(1, 2) = A(0, A(1, 1)) = A(0, 3) = 4$.

From (5): $A(1, 3) = A(0, 4) = 5 \Rightarrow A(2, 1) = 5$.

Finally, from (1): $A(2, 2) = A(1, 5)$.

Compute $A(1, 5) = A(0, A(1, 4))$, $A(1, 4) = A(0, A(1, 3)) = A(0, 5) = 6$.

Thus $A(1, 5) = A(0, 6) = 7 \Rightarrow \boxed{A(2, 2) = 7}$.

Chapter 2

Previous Year Questions

Demonstrate the maximum number of regions that can be formed by n zig-zag lines. Calculate the maximum number of regions that can be obtained by ZZ_n ,

[6]

F-24

$n = (\text{last digit of your ID}) + 1$, If the last digit of your ID < 9

$n = 3$ If the last digit of your ID $= 9$

Prove $J(2^m + L) = 2l + 1$ for the Josephus Problem using the mathematical induction method where $n = 2^m + L$. Also, find $J(103)$ using the recurrence method.

[6]

F-24

Compare Googol and Googolplex. Then Generate the value of 22 by categorizing each step of Ackermann functions.

Two fair 12-sided dice are rolled. Let X be the sum of the numbers appearing on the two dice. Develop the probability that the sum is at most 6 ($X \leq 6$).